Automated Semantic Extraction Pipeline

Objective and overview

This pipeline automates the part of the project that was performed manually on 9 questionnaire items (PHQ-9, GAD-7, Epworth Sleepiness Scale): linking each item to a SNOMED CT concept, assessing the quality of the correspondence (Exact / Broader / Narrower / No Match), then generating an RDF/Turtle fragment.

The pipeline consists of four steps, each in its own script, with a separate output file between each step:

1. extraction_qwen_automatisee.py — a local LLM (Qwen2.5-7B-Instruct, via Ollama) proposes a clinical concept for each item.

2. recherche_snomed_B2.py — the BioPortal search API returns real SNOMED CT concepts that are candidates for this term.

3. decision_match_C2.py — the LLM chooses, from this closed list of candidates, the one that best corresponds (or answers No Match).

4. generation_ttl_D2.py — a script automatically generates the RDF/Turtle fragment from this decision.

The purpose is to compare an automated result, in order to measure what an LLM actually contributes compared with a conventional terminology search.

The 9 items processed

The items were selected to cover varied semantic categories rather than testing the pipeline only on affective symptoms.

Questionnaire	Items	Why this choice
PHQ-9	phq2, phq4, phq7, phq9	mood, somatic symptom, cognition, safety item (suicidal ideation)
GAD-7	gad1, gad4, gad6	general anxiety, psychophysiological manifestation, irritability
Epworth Sleepiness Scale	epw4, epw8	two situations with different alertness load (passive passenger vs stopping in traffic)

Role of each file

None of the scripts is allowed to invent a SNOMED value: B2 retains only identifiers returned as such by the API.

Number of SNOMED candidates searched

B2 requests 10 candidates per concept from the BioPortal API (PAGESIZE = 10).

This choice is an empirically observed compromise: with 5 results, the correct concept was sometimes missing from the list (cases of Depressed mood or Suicidal thoughts, absent from the top 5 for the searched terms). With 10, the correct concept appears more often, but the longer list also introduced new errors: for one item, the LLM incorrectly chose a disorder-type concept (Mild depression) that appeared only because of the expansion.

Expanding the candidate list therefore improves recall but does not automatically improve the precision of the final choice.

File	Role	Input	Output
extraction_qwen_automatisee.py (A2)	Asks the LLM for a clinical concept per item	the 9 items (coded in the script)	resultats_A2.json / .csv
recherche_snomed_B2.py (B2)	Queries the BioPortal API to find real SNOMED concepts	resultats_A2.json	resultats_B2_candidats.csv
decision_match_C2.py (C2)	Asks the LLM to choose a candidate or No Match	resultats_A2.json + resultats_B2_candidats.csv	resultats_C2_match.csv
generation_ttl_D2.py (D2)	Builds the RDF/Turtle fragment from the decision	resultats_A2.json + resultats_C2_match.csv	questionnaire_ontology_AUTO_D2.ttl
valider_ttl_D2.py	Checks the .ttl syntax and the consistency of owl:sameAs	the .ttl + resultats_C2_match.csv	a report displayed in the console

Instructions given to the match decision prompt (C2)

The LLM receives the item and the closed list of candidates returned by B2 (identifier, preferred label, definition, synonyms), and must choose the best candidate or answer No Match, with a short justification.

Definitions imposed in the prompt:

Exact: the candidate and the item's construct have the same scope.

Broader: the candidate concerns the same clinical construct but in a more general way.

Narrower: the candidate concerns the same clinical construct but in a more specific way.

No Match: no candidate concerns the same clinical construct (different clinical domain).

A priority rule specifies that, as soon as a candidate concerns the same construct, even partially, Exact/Broader/Narrower must be chosen — No Match is reserved for cases involving a completely different clinical domain. A second rule specifies that the “construct” covers only the clinical content, not the response format (frequency, situation): a candidate must never be rejected simply because it does not reproduce the item's scale or situational context.

Three anti-hallucination safeguards are applied by the SCRIPT, not only by the prompt: an identifier returned that does not belong to the candidate list automatically forces No Match; an item with no candidates is declared No Match without even calling the LLM; the final retained label always comes from B2, never from what the LLM wrote.

Results obtained

Item	Manual concept	Manual match	Automatic concept	Automatic match	Agreement
phq2	Depressed mood	Narrower	Mild depression	Narrower	Different concept
phq4	Fatigue	Exact	Fatigue	Exact	Full agreement
phq7	Poor concentration	Exact	ADHD, combined type	Narrower	Different concept (inferred diagnosis)
phq9	Suicidal thoughts	Narrower	Suicidal thoughts	Exact	Same concept, match more severe manually
gad1	Anxiety	Narrower	Anxiety	Exact	Same concept, match more severe manually
gad4	None	No Match	None	No Match	Full agreement
gad6	Feeling irritable	Exact	Feeling irritable	Exact	Full agreement
epw4	Excessive daytime sleepiness	Narrower	None	No Match	False negative
epw8	Excessive daytime sleepiness	Narrower	None	No Match	False negative

Across the 9 items: 3 full agreements, 2 partial agreements (same concept, severity judged differently), 2 disagreements on the selected concept, 2 false negatives.

Known limitations

Errors persist despite explicit prompt corrections. For epw4/epw8, the correct candidate (Excessive daytime sleepiness) was presented in first position and was still rejected, even after adding a dedicated rule to the C2 prompt. The 7B model does not always follow a nuanced instruction reliably.

B2 depends entirely on the quality of the term proposed by A2. B2 only searches for the word that the LLM itself produced in A2; if this term is imprecise (“depression” instead of “depressed mood”, “attention_deficit” instead of “poor concentration”), the correct SNOMED concept sometimes never appears among the candidates proposed to C2, regardless of the quality of the C2 prompt afterward.

Expanding the candidate list can introduce new errors. Moving from 5 to 10 candidates made it possible to retrieve some missing concepts, but also caused an erroneous choice to appear (a disorder-type concept) that did not exist with a shorter list.

Composite items remain a structural problem. When an item combines several notions (“A, B, or C”), the automated pipeline tends to be less cautious than manual validation when choosing between Exact and Narrower.